CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 3: Greedy Approach Huffman & Spanning Trees

Course: Design Analysis and Algorithms [23CS3502]
Category: Assignments | Assignment #3 | Academic Year: 2026-27

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1. Overview & Objective:

Formulation of greedy-choice property and optimal substructure guarantees.

2. Assignment Problems & Instructions:

1. Construct a Huffman Tree for given character frequency table and compute bits saved.
2. Formulate and prove optimal solution for the Fractional Knapsack problem.
3. Compare Prim algorithm with Kruskal algorithm using Disjoint Set Union data structure.
4. Solve Activity Selection problem and prove greedy strategy produces maximum activities.

Submission Guidelines & Verification:

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